

Invasion percolation Markov Chains – A probabilistic framework for assessing vertical CO₂ migration

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ARTICLE INFO

Keywords:

CO₂ storage
Vertical CO₂ migration
Containment risks
Leakage
Invasion percolation
Markov
Stochastic model

ABSTRACT

Potential CO₂ storage sites need to perform risk assessments on the likelihood of anomalous events such as leakage. The intrinsic heterogeneity of the rock system with uncertain values for the capillary threshold pressures of the various rock elements is the most likely reason for unexpected vertical migration of CO₂ within a storage complex. This study shows how the Invasion Percolation Markov Chain approach can be used to address this concern. We tested the approach using detailed 3D models of the multi-layer plume at Sleipner showing that even small variations in the threshold pressures of the shales can impact the flow of CO₂ into multiple accumulations. Models with and without shale breaks reveal the importance of vertical feeders and/or faults, and the geometry of the shale layers is also crucial as the CO₂ strongly conforms to topography. We demonstrate that the vertical migration of CO₂ at Sleipner follows a Markovian model in which the probability of later migration events is highly dependent of the probability of preceding events. This case study illustrates how the initial migration events, which have the highest probability of occurring, should be the focus of CO₂ storage risk assessments.

1. Introduction

CO₂ storage refers to the engineered geological storage of CO₂ which takes large volumes of CO₂, that would otherwise have been emitted into the atmosphere, and stores them in the deep subsurface, isolating it for thousands to millions of years. There are numerous geological storage options for CO₂ including saline aquifer formations, depleted oil and gas reservoirs, coal beds, and mafic/ultramafic rock formations. In this paper we focus on saline aquifers. Captured CO₂ is usually injected in the subsurface at depths exceeding 800 m to ensure it remains in a dense form which enhances storage efficiency (Ringrose, 2020). Additionally, storage at greater depths improves storage security, as sealing formations become more effective due to increased compaction and cementation (Marcussen et al., 2010). CO₂ properties in the subsurface are highly variable, depending on pressure and temperature, and CO₂ can enter the gas phase during migration to shallower units or leakage to the surface. CO₂ is less dense than the resident brine (saline aqueous phase), so the mobile CO₂ plume tends to migrate up dip and fill structural traps leaving some residual CO₂ behind due to pore-scale trapping. CO₂-brine

reactions are also expected, such as salt precipitation near the injector, if the CO₂ stream has low aqueous content, and dissolution of the CO₂ into the brine phase, referred to as solubility trapping. Dissolution in the brine and mineralisation reactions will contribute to immobilisation of the CO₂ in the long-term (Gilmore et al., 2020; Ringrose et al., 2021).

A CO₂ storage monitoring programme needs to be defined before the start of injection and covering the whole lifetime of a project. The primary objectives of monitoring are to demonstrate conformance (Oldenburg, 2018), showing that the CO₂ plume behaves as expected, and to ensure containment. If anomalous events are detected, response actions need to be defined in a contingency plan. Potential leakage out of a storage complex could happen vertically into the overburden or laterally, beyond a license boundary for example. Vertical migration of CO₂ out of a geological storage unit could occur along wellbore damage zones, faults/fractures or via complex pathways in the overburden. If CO₂ flows out of the storage complex and upwards into the overburden complex, the flow paths can be complex and there will eventually be a phase transition from liquid to vapor. Migration of a buoyant non-wetting liquid/gas phase is likely to occur as percolating clusters, approximated by invasion percolation (IP) theory. Continuous flow of

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<https://doi.org/10.1016/j.ijggc.2025.104338>

Received 24 September 2024; Received in revised form 5 February 2025; Accepted 22 February 2025

Available online 3 March 2025

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Nomenclature

C(i)	capillary threshold pressure function (for rock unit i)
h	fluid column height [m]
P	fluid pressure [Pa]
Pr	probability
P_{th}	threshold pressure [Pa]
V	volume [m ³]
V_{ri}	residual volume [m ³]
T, t	thickness of geological units [m]
ϵ	indicator variable
γ	interfacial tension [mN/m]
μ	mean
σ	standard deviation

CO₂ along faults or well-bore damage zones is more likely to be a Darcy flow process (e.g. Snippe et al., 2022).

Prospective CO₂ storage sites need to perform leakage risk assessments (Pawar et al., 2015) to analyse the likelihood of these anomalous events and to define a monitoring plan to help ensure that migration out of the storage unit into high-risk rock elements such as faults can be detected (Romanak et al., 2012). Risk assessments will require a probabilistic component due to the uncertainties in determining the properties of rock systems (e.g. Pawar et al., 2015). The focus of this paper is on assessing the uncertainties associated with vertical CO₂ migration in a storage complex using a stochastic model for this phenomenon. Based on IP theory (Wilkinson and Willemsen, 1983; Frette et al., 1992), we model the movement of CO₂ volumes in different rock units by a Markovian process. The probabilities required in this model description will rely on geological and geophysical understanding of the formations' geometry and the capillary threshold pressures in relevant rock units. The Markov assumption quantifies the absence or presence of CO₂ in one rock unit, given what occurs in the deeper units, and depends only on the outcome of the unit directly below. A version of this approach is demonstrated in Callioli Santi et al. (2022). In the current work we extend that approach with a particular focus on results and interpretations for the Sleipner CO₂ Capture and Storage (CCS) project, using it as a well-constrained case study to inform future storage projects.

The Sleipner CCS project has been in operation since 1996 and has been monitored by time-lapse seismic surveys every 2 to 4 years (e.g. Arts et al., 2004; Eiken et al., 2011; Furre et al., 2017; Furre et al., 2024) which have demonstrated safe storage and provided excellent imaging of the CO₂ plume development and distribution. Therefore, it provides valuable insights on the CO₂ plume development in a multi-layer storage unit, which can be used as a proxy for vertical CO₂ migration in multi-layered stratigraphic systems in general. The CO₂ plume is differentiated into nine distinct layers within the Utsira Formation storage unit, with complex vertical CO₂ migration patterns caused by backfilling of CO₂ below thin intraformational mudstones and by vertical corridors caused by faults or breaks within the mudstones (Furre et al., 2024). These intra-formational shales are an important component of the storage process at Sleipner, with less than 20 % of the CO₂ reaching the caprock after 14 years of injection (discussed further below). Bump et al. (2023) introduced the concept of composite confining systems in which discontinuous barriers in multi-layered stratigraphic systems provide effective containment as small capillary entry pressures are enough to divert flow and restrain vertical migration of CO₂, depending on barrier frequency and area. Therefore, the immobilisation of CO₂ in these systems is enhanced by CO₂ migration. They mention that the Sleipner case illustrates the advantages and risks of such systems with the intraformational barriers limiting the lateral spread of the CO₂ plume, pointing out that the alignment of the breaks in

these barriers (holes and/or pervasive fractures) have reduced the potential effectiveness of the system since the CO₂ plume reached the base of the Nordland shale after only 3 years of injection.

There are numerous studies on fluid flow simulation models calibrated to the seismic observations at Sleipner (e.g. Singh et al., 2010; Chadwick et al., 2010; Chadwick and Noy, 2010; Cavanagh et al., 2015; Williams and Chadwick, 2017), which have significantly improved our understanding of the behaviour of a CO₂ plume in the subsurface and demonstrated that gravity forces are an important control on the plume behaviour (Ringrose, 2020). Most of these studies have focused on history matching the topmost layer 9. Matching the complete nine-layer plume distribution in Sleipner is still a challenge due to poorer seismic imaging of deeper layers, uncertainties in the geology (particularly flow pathways within the thin shale baffles) and fundamental aspects of the plume dynamics (including the plume temperature profile and uncertainties in estimating the fluid saturation; Cavanagh and Haszeldine, 2014). The IP method has the advantage of being able to rapidly compute flow pathways for a non-wetting fluid invading a complex (water wet) porous medium. Although it neglects viscous forces, it can be a good approximation for flow problems where the flow physics is expected to be dominated by gravity and capillary forces.

Here we use an IP simulation approach and apply it to the Sleipner nine-layer 3D reservoir model (using the dataset available on the CO₂ DataShare portal) to illustrate how uncertainties in the threshold pressures of the thin intraformational shale layers and the introduction of breaks in these layers (vertical conduits) can impact the probability of the vertical migration of CO₂ from the injection point to the caprock.

The IP simulation approach quantifies the process where CO₂ is trapped in structural closures due to capillary forces acting at the pore scale and controlled by the interfacial tension at fluid interfaces. The percolating clusters of CO₂ accumulate underneath low permeability barriers (e.g. shale layers) until they spill laterally or breach the barriers when the buoyancy pressure overcomes the threshold pressure. Note that the vertical migration defining the nine CO₂ layers is constrained to the Utsira Fm. storage unit (not the overburden). The Sleipner dataset thus displays a well contained multi-layer CO₂ plume with no indication of any CO₂ migrating into the overburden, based on a total of 10 seismic repeat surveys (Furre et al., 2024).

Our sensitivity analysis of the threshold pressures of the shales consisted of a total of 500 realisations for each scenario, i.e. a laterally continuous shales scenario (CS) and 'shales with breaks or vertical conduits' scenario (SB), in which the threshold pressures of each shale layer were allowed to vary independently from each other before running IP simulation for each realisation. The CO₂ mass distribution (Mt) per layer in 2010 for the 500 realisations of threshold pressures for each scenario (CS and BS) gave us insights on the probability distribution of CO₂ mass per layer (500 realisations were chosen because this number reasonably spans the dimensions of input variables in our setting and is significantly higher than the 100 realisations typically used in ensemble-based geostatistical simulations). We analysed the accumulation frequency per layer for all 500 realisations per scenario in each timestep which indicated how fast the CO₂ reached the topmost L9 and the impact of time on the probability of vertical migration of CO₂.

We have structured this paper so that we start with the description of vertical migration of CO₂ based on IP concepts in Section 2. In Section 3 the Invasion Percolation Markov Chain method is first illustrated in a conceptual model of a multi-layer stratigraphic system (in Section 3.1) and then applied to the Sleipner nine-layer 3D model using an IP simulation approach (in Section 3.2). In Section 4 we present the results from the application to the Sleipner nine-layer 3D model, which are then discussed in Section 5, with our overall conclusions given in Section 6.

2. IP description of vertical CO₂ migration

Buoyancy and capillary pressures are the two competing forces controlling hydrocarbon migration and entrapment in the subsurface.

Buoyancy drives less dense fluids upwards while capillary pressure forces denser fluids to move downwards. When buoyancy pressure exceeds the capillary pressure, the brine will be displaced and the pore spaces will become occupied by hydrocarbons or CO₂ (giving $S_w < 100\%$) (Slatt, 2006). Buoyancy forces increase with the density difference between pore water and hydrocarbon and with increasing height of the hydrocarbon column, such that:

$$P_c = \Delta\rho gh \quad (1)$$

where, P_c is the capillary pressure of the invading phase at the interface, $\Delta\rho$ is the density difference between the fluid phases (such as CO₂ and brine), g is the standard gravity, h is the column height of the invading phase (for instance CO₂).

The macroscale phenomenon of capillary pressure results from the interaction of forces acting between the fluid phases and the solid (rock grains) at the pore scale. These include cohesive (fluid-fluid) and adhesive forces (fluid-solid) (Vavra et al., 1992). The fluid is defined as the wetting fluid if adhesive forces are greater than cohesive forces (i.e. it spreads out on the pore walls). If cohesive forces are higher, the fluid is called non-wetting. The relative wettability of the fluids is given by the contact angle θ (angle between the solid and the fluid-fluid interface measured through the denser fluid). CO₂ driven into a saline aquifer defines a two-phase immiscible flow system and water is typically the wetting fluid (Nordbotten and Celia, 2011). In our study, we assume a water-wet system with $\cos\theta = 1$. The pore-scale interfaces between the fluid phases and the solid are controlled by the interfacial tension (γ) in which each fluid phase pulls to minimise its boundaries. This is a function of density difference with a higher density contrast between two fluids increasing γ . Therefore, it is also dependent on pressure, temperature and brine salinity as they impact the densities of the reservoir fluids. Naylor et al. (2011) have shown that in CO₂-brine systems γ generally increases with decreasing pressure hence the vertical migration of CO₂ will tend to encounter increasing resistance due to the increasing γ . By definition, capillary pressure is the difference in pressure between the non-wetting phase and the wetting phase. It is inversely related to the pore size, hence the larger the pore the smaller the capillary entry pressure (or threshold pressure) required for the non-wetting fluid to displace the wetting phase. This can be expressed mathematically as:

$$P_{th} = \frac{2\gamma(\cos\theta)}{r_c} \quad (2)$$

where, P_{th} is the threshold pressure or the amount of pressure required to force the non-wetting phase (oil, gas, CO₂) to displace the wetting phase (brine), γ is the interfacial tension between the fluids (such as CO₂ and brine), θ is the contact angle, r_c is the capillary (pore) radius (Slatt, 2006).

There is a significant contrast in density between the injected CO₂ and the resident brine in a storage reservoir which creates strong buoyancy forces driving the mobile CO₂ plume upwards. The plume movement is impeded by the capillary entry pressure (or threshold pressure) of less permeable layers (baffles or barriers) causing the mobile CO₂ to accumulate in high permeability layers filling structural or stratigraphic traps and spreading laterally until the buoyancy pressure of the CO₂ accumulation exceeds the threshold pressure of the barrier or the CO₂ finds a high permeability vertical pathway. This process creates multiple accumulations of CO₂ (as demonstrated by Trevisan et al., 2017 using laboratory experiments). Therefore, the CO₂ plume is strongly impacted by formation heterogeneities.

The injection of dense or supercritical CO₂ increases the pressures near the well forcing it to enter the pore spaces initially occupied by formation fluids. As the CO₂ flows away from the well it gradually transitions from viscous to capillary and gravity dominated flow (Ringrose, 2020). The dimensionless capillary number (C_a) can be used to describe the limits between viscous and capillary flow (Cavanagh

et al., 2015) which is estimated as follows:

$$C_a = \frac{\mu q}{\gamma} \quad (3)$$

where, μ is the viscosity, q is the flow rate and γ is the interfacial tension. Capillary numbers around 10^{-4} or less define the boundary condition for buoyant fluids to percolate through the rock formations as gravity-driven ganglia (England et al., 1987). Cavanagh and Haszeldine (2014) estimated a capillary number of about 10^{-7} for the Sleipner CO₂ plume implying that the flow can be approximated by IP.

IP theory describes the behaviour of immiscible fluids (such as CO₂) migrating or percolating into porous medium as it displaces the original pore fluid (brine) (Wilkinson and Willemsen, 1983; Frette et al., 1992) and has been successfully applied in regional hydrocarbon migration and hydrocarbon field charge process (Carruthers and Ringrose, 1998). It assumes a capillary and gravity dominated flow domain, which is generally expected as the CO₂ moves into the far-field region away from the pressure gradients close to injection wells and in the long-term as gravity-capillary equilibrium is reached (Ringrose et al., 2021).

As CO₂ migrates through a formation, a drainage process occurs at the front of the CO₂ plume (CO₂ displaces brine which is trapped at irreducible water saturation), whereas an imbibition process takes place along the path of the plume (where water imbibes back into the domain and CO₂ is trapped as residual saturation). The residual trapping of CO₂ along the path of the plume depends on rock heterogeneities (such as pore throat sizes, interfacial tension and wettability) and injection strategy. Therefore, a certain volume of CO₂ is left behind the migrating plume which can be related to residual saturations in the range of 20–30 % (based on laboratory measurements; Krevor et al., 2012). Over time, much of the trapped CO₂ dissolves into the formation water.

Based on these concepts, an Invasion Percolation Markov Chain (IPMC) model is proposed where the vertical migration of CO₂ from a permeable rock element (t_i) through a low permeability layer (T_i) occurs when the maximum fluid pressure (P_{vi}) of an accumulated volume of the buoyant non-wetting fluid in permeable rock unit t_i (V_{ti}) is greater than the capillary threshold pressure of low permeability rock unit T_i (i.e. $P_{vi} > C(T_i)$). The maximum fluid pressure (P_{vi}) depends on the column height of the accumulation (h_i) and the fluid density contrast between CO₂ and brine ($\delta\rho = (\rho_{br} - \rho_{CO_2})$). P_{vi} is non-linearly dependent on the geometry and fluid density of the accumulated volume V_{ti} , i.e. $P_{vi} = f(h_i, \delta\rho, V_{ti})$. Therefore, the IP process, after an initial migration event, continues upwards through the subsequent rock element pair if the following condition is met:

$$P(V_{ti} - V_{ri}) > C(T_i) \quad (4)$$

where P represents the buoyancy pressure of the accumulation of CO₂; t_i corresponds to permeable rock element, $i=1, \dots, m$; T_i to a low permeability layer, $i=1, \dots, m$; V_{ti} is the volume of the non-wetting fluid in permeable rock element t_i ; V_{ri} is the residual volume; $C(T_i)$ is the threshold pressure of the low permeability layer T_i . The Nomenclature section summarises the main notations used in this paper. Therefore, P corresponds to the remaining volume after the residual volume V_{ri} is trapped on the path of the plume in the pore system of the permeable rock element t_i . Then a certain fraction of volume percolates through the capillary barrier and invades the permeable layer above. Hence, in a storage system with multiple geological barriers (low permeability layers) the vertical migration of CO₂ will have a strong tendency to cease due to the gradual loss of the remaining volumes V_{ti} . Fig. 1 shows an illustration of this concept where the probability of a vertical migration event from permeable rock elements t_1 to t_2 is given by $Pr(P(V_{t1} - V_{r1}) > C(T_1))$. It also illustrates other possible pathways that the CO₂ could preferentially follow due to lower capillary threshold pressures such as faults (ϵ_{1f}) and sand injectites (ϵ_{1i}).

The main factor behind unexpected vertical migration of CO₂ in such a rock system is inaccurate estimations of capillary threshold pressures

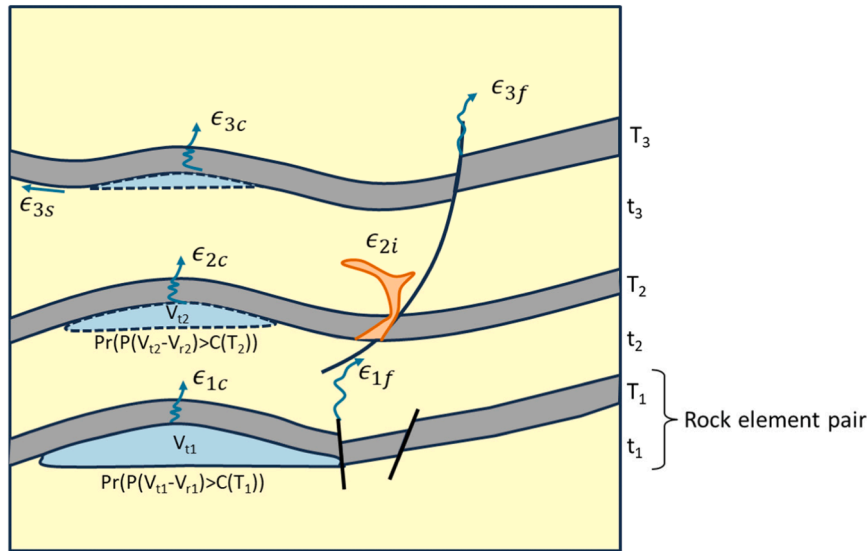


Fig. 1. Simplified sketch of rock-element pairs (t_i , T_i) containing a volume, V_{ti} , of a buoyant non-wetting fluid (CO_2). Migration events could occur when the fluid buoyancy pressure P_{v1} exceeds the capillary threshold pressure of the sealing layer, $C(T_1)$, as with event ϵ_{1c} , or they could occur by finding lower capillary-threshold pressures pathways at faults (as for events ϵ_{1f} and ϵ_{3f}) or along injection features (such as in ϵ_{2i}), or by lateral migration via spill points (such as in ϵ_{3s}).

of the rock elements due to the inherent heterogeneity of the rock system (Callioli Santi et al., 2022). This highlights the need for detailed geological characterisation of the storage complex (including overburden) and a framework to handle their associated uncertainties and corresponding potential for vertical migration of CO_2 .

3. Stochastic modelling of vertical CO_2 migration

Considering the IP framework, the vertical CO_2 migration process can be modelled as a Markovian process under some assumptions. Note that in a CO_2 migration setting, the Markov transitions are of a special type in the sense that migration to one element can only occur if there is already sufficient CO_2 in the element immediately below. In absence of CO_2 in the element below, there can of course be no migration (assuming the first-order Markov model for vertical migration, without any fault or other vertical pathways). Martinelli et al. (2011) applied similar Markovian ideas in a Bayesian network model for hydrocarbon prospects source rock modelling, while Larsen et al. (2006) imposed such vertical ordering of saturations to enhance lithology and fluid prediction from seismic data. In this study, we adopt a Markov model for assessing the presence or absence of CO_2 in a multi-layered stratigraphic system.

3.1. Conceptual model

We first illustrate an elementary model for the statistical volume distribution in different geological storage elements, assuming CO_2 is injected in the bottom element of the m stratigraphic columns. The accumulated CO_2 volume in an element is:

$$V_{ti} \geq 0, \quad i = 1, \dots, m, \quad (5)$$

where outcome 0 indicates no accumulation of CO_2 , i.e. no CO_2 has migrated into this element. The statistical model for the volumes will be a mixture distribution with a probability mass at 0 and a continuous probability density for higher volumes. The (initial) probability of volumes in the lowermost element is $\Pr(V_{t1} > 0) = 1$, because of the direct injection taking place in this element.

Next, the Markovian transition model for CO_2 volumes is described as follows:

$$\Pr(V_{ti} = 0 | V_{ti-1} = 0, \dots, V_{t1}) = \Pr(V_{ti} = 0 | V_{ti-1} = 0) = 1, \quad i = 2, \dots, m \quad (6)$$

$$\Pr(V_{ti} = 0 | V_{ti-1} > 0, \dots, V_{t1}) = \Pr(V_{ti} = 0 | V_{ti-1} > 0) < 1, \quad i = 2, \dots, m. \quad (7)$$

Here, Eq. (6) states that migration cannot occur without any volume accumulation in the below element, while Eq. (7) describes the chance of migration when an accumulation exists in the below element. Using the Markov assumption one here removes the conditioning to elements other than that immediately below. This is valid under the assumption of known geometry and capillary threshold pressures in elements $i - 2, \dots, 1$ so that residual volumes in the deeper layers can be assumed known, which is done computationally with stepwise simulation of such IP processes. In Eq. (7), the continuous volume outcomes $V_{ti} > 0$ are represented by a probability density function. Such distributions for accumulated volumes of element i are illustrated in Fig. 2. In practice, the density function is implicitly defined via the distributions for geometry and the capillary threshold pressure for element $i - 1$. The specified input probability distributions for capillary properties and geometries must be realistic and aligned with the physical conditions controlling migration in the IP framework, using distributions for each physical parameter in every unit. Such realistic simulations of IP mechanisms have been previously conducted by e.g. Oldenburg et al. (2016), Tavassoli et al. (2021) and Callioli Santi et al. (2022). There are also various software tools that can be applied for this task such as the Permedia® CO_2 software which is used in this study.

We can then mimic the migration behaviour and its uncertainty using Monte Carlo sampling. A realisation is generated by sampling geometries and capillary threshold pressures for each element. The IP migration process will then proceed when the accumulated pressure exceeds the limits which the capillary threshold pressure can handle; $P(V_{ti} - V_{ri}) > C(T_i)$ as demonstrated in Section 2. The Monte Carlo simulation then proceeds element by element over the migration times for the set injection rates over time. By running multiple Monte Carlo realisations of this migration process, we can gain understanding about the behaviour of the injected CO_2 over time as demonstrated in the next section.

3.2. Application to the Sleipner nine-layer 3D model

The Sleipner CO_2 storage project in the Norwegian North Sea is used as a case study for demonstration of the method. Seismic time-lapse monitoring has provided excellent imaging of the CO_2 vertical

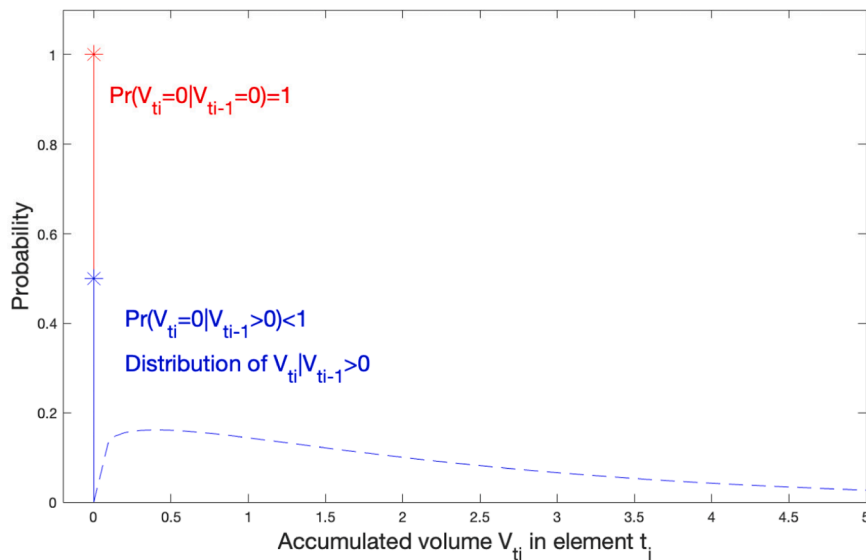


Fig. 2. Illustration of the distribution for CO₂ volume accumulation in an element. Knowing that no CO₂ has accumulated in the element below, there is for sure 0 accumulation in this element. Knowing that the CO₂ has started to accumulate in the element below, there is a chance of migration giving accumulation into this element, with uncertain values illustrated by the density function.

migration patterns through the multi-layer Utsira sandstone storage unit (Fig. 3). Even though Sleipner is not a leakage case, it provides valuable insights into the nature of CO₂ vertical migration patterns through partially sealing intraformational shales which are relevant to vertical migration of CO₂ through caprock seals out of a storage unit.

The Utsira sand unit at Sleipner contains thin spikes of high gamma-ray and density well log values (similar to the overlying Nordland shale), but with the wide separation of wells (>1 km) correlation is difficult, and they are not possible to observe in the baseline seismic due to their thickness of about 1 m. In the early stages of the Sleipner CO₂ storage project these thin shales were expected to constrain the CO₂ vertical migration but not to be completely sealing (Zweigel et al., 2004), which was confirmed by the 1999 3D time-lapse seismic survey. Based on pre-injection data (wells and baseline seismic) and the first-time lapse

seismic, it was then possible to interpret that the upward migration of CO₂ was restricted by at least 4 internal shale layers as illustrated in the sketch proposed by Zweigel et al. (2004) (shown in Fig. 4).

The IPMC method can be applied analytically for a simple case, as done by Callioli Santi et al. (2022) for the *a priori* knowledge set for the Sleipner case which assumed four intra-Utsira laterally continuous shale layers. For this early-stage model, a simple ellipsoidal cap model was assumed along with likely threshold pressures, as illustrated in Fig. 4. Using a maximum column height of 30m for the deepest layer and applying the same threshold pressure distribution for each shale layer, the resulting buoyancy force for this layer was much lower than the assumed mean shale properties (~120 kPa vs P_{th} mean of 764 kPa with corresponding 0.001 mD mean permeability). This simple model gave a migration probability from the deepest layer to the layer immediately above of about 5 %. Due to residual trapping (assumed as 20 %), the maximum column height decreases as CO₂ migrates upwards hence the probability of the CO₂ accumulations exceeding the threshold pressures of the shale layers and migrating vertically decreases. The estimated joint probability of migration out of the topmost layer was 0.03 %.

Next, to test the IPMC concept for a more mature model dataset, the Sleipner 2019 Benchmark 3D model (available on the CO₂ Datashare portal) was used in an IP simulation approach with nine sandstone layers (L9 to L1 from shallowest to deepest layer) vertically separated by thin laterally continuous shale layers (thickness of about 1.5 m with the exception of the Thick Shale that separates the upper layers L8 and L9 at approximately 7.5 m thick). This model is based on baseline seismic interpretations (1994) and time-lapse seismic CO₂ injection image data up to 2010 with significant uncertainties from the time-lapse seismic data especially in the deeper layers L4 to L1 (as documented in the CO₂ Datashare portal). The 3D grid has a lateral extent of 3.2 km by 5.9 km with lateral resolution of 50 m × 50 m. It was built using a proportional layering scheme with vertical resolution of approximately 0.5 m in the intrashale layers, 2 m in the sandstone zones and 5 m in the caprock, with a total of about 2 million cells (64 × 118 × 263 cells in I × J × K).

The CO₂ at Sleipner is injected into the lower part of the Miocene-Pliocene Utsira Fm. comprising highly porous and permeable weakly consolidated marine sandstones that are under hydrostatic pressure conditions (Singh et al., 2010). The base of the Utsira Fm. defines an unconformity with presence of multiple mud edifices caused by local mobilisation of the underlying Hordaland shale which results in significant local thickness variations across the reservoir due to their

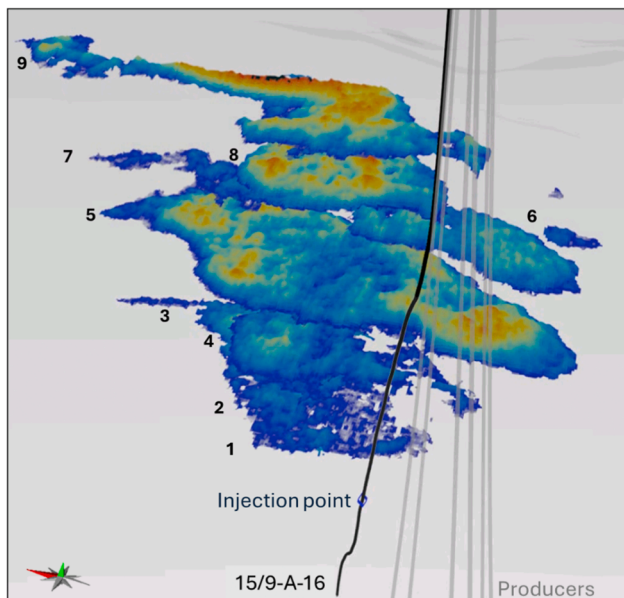


Fig. 3. 3D reconstruction of the CO₂ plume at Sleipner at the time of the 2008 time-lapse survey, showing the multi-layer CO₂ distribution (compass bottom left: red points north and green points up). Based on work by Kier (2015), image courtesy of Equinor.

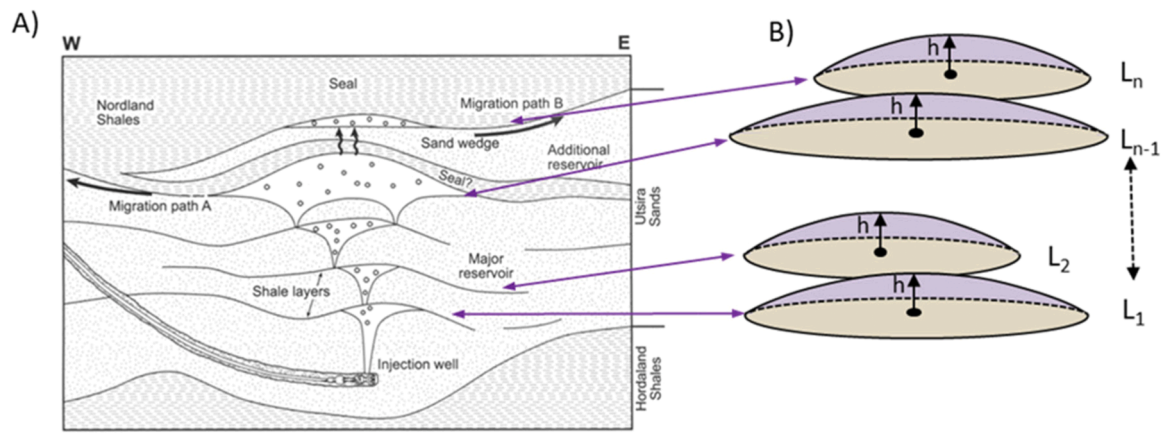


Fig. 4. . Illustration of the analytical approach. A) [Zweigel et al. \(2004\)](#) proposed concept of likely upward migration of CO₂ in the Utsira Fm. based on data available at and early phase of the project. B) 4-layer stacked ellipsoidal caps model ([Callioli Santi et al., 2022](#)).

preferential compaction during burial ([Zweigel et al., 2004](#)). There is limited acquisition of core data of the Utsira sandstone with many interpretations for their depositional environment ([Rundberg, 1989](#); [Gegersen et al., 1997](#); [Eidvin et al., 1999](#); [Zweigel et al., 2004](#); [Williams and Chadwick, 2021](#); [Furre et al., 2024](#)). [Furre et al. \(2024\)](#) discuss recent data used to monitor the site and argue that the sandstone units were deposited in a relatively deep-water setting as submarine fan deposits, in contrast to earlier interpretations which suggested a shallow marine depositional environment.

The main inputs and assumptions used in the IP simulation approach are summarised in [Appendix A](#). In the IP simulations for the nine-layer Sleipner case the trapping mechanisms considered are structural, stratigraphic and residual. Hence, CO₂ dissolution in the brine phase and mineralisation are not accounted for in this study but they would also contribute to permanently immobilise the CO₂. The layered CO₂ plume distribution caused by the presence of the thin mudstones in the Utsira Fm. maximises the contact surface between the two phases which enhances the dissolution of CO₂ into the brine, with estimates for Sleipner of 0.85 % per year based on reservoir simulations ([Nazarian and Furre, 2022](#)) and an upper bound of 1.8 % per year based on gravimetry data ([Alnes et al., 2011](#)).

Considering the rock and fluid property assumptions (described in [Appendix A](#)) and the Sleipner 2019 Benchmark 3D grid structural setting, the percolating flow of CO₂ will therefore be dependent on the CO₂ capillary pressures and shale threshold pressures. Laboratory measurements from samples of the caprock from production well 15/9A-11 close to the storage site indicated threshold pressures of about 1.7 MPa ([Springer and Lindgren, 2006](#); [Singh et al., 2010](#)) which if applied to the intraformational shales of Sleipner results in all the CO₂ accumulating and laterally spilling beneath the lowermost barrier L1 ([Callioli Santi, 2019](#)). In order for the CO₂ to break through the intraformational shale layers and form the observed vertical stack distribution of nine layers, their threshold pressures need to be significantly smaller than the sampled caprock values, as demonstrated by [Cavanagh et al. \(2015\)](#).

A deterministic assessment was initially performed to determine the threshold pressures required for the laterally continuous intraformational shales to act as baffles, allowing the replication of the vertical stack of nine CO₂ plume layers in 2010 (total of 12.18 Mt of CO₂ injected). The starting point for this analysis was the modelled threshold pressures values of 38 to 56 kPa defined in [Cavanagh et al. \(2015\)](#). These values were set as constants per corresponding intraformational shale and resulted in about 95 % of the total injected mass in 2010 accumulating in the topmost L9. Hence, they were systematically increased to enhance the distribution of mass in the lower layers' accumulations L1 to L8. Property modelling using Sequential Gaussian simulation with normal distributions for the porosity and threshold pressures was

applied. Note that permeability is not a direct rock property input in the IP simulation approach but is correlated to the threshold pressure. Assigning a normal distribution to the threshold pressures of the intra-shale layers ($\mu = 98$ kPa and $\sigma = 16$ kPa) resulted in some CO₂ accumulating beneath each intraformational shale, and then the total mass of CO₂ injected by 2010 accumulating in the uppermost L9 was reduced to 45 %, which defined the deterministic base case. The threshold pressures values reported in this study correspond to CO₂-brine system. We assumed a γ value of 0.0625 ([Singh et al., 2010](#)) when converting Mercury-Air (Hg-Air) to CO₂-brine system, since the input values into the Permedia® software are in Hg-Air system ([Appendix A](#)). However, since γ increases with decreasing depth, the same threshold pressure distribution to all shale layers in Hg-Air system will increase slightly from the deep layer 1 to the shallow L9 once converted to reservoir conditions.

The simulation results showed a large sensitivity to small variations of the threshold pressures of the shales in this deterministic analysis. To evaluate possible distributions of threshold pressures in the laterally continuous shales, a stochastic assessment was then performed defining a sensitivity analysis of the threshold pressures of the shales with a total of 500 IP simulation runs (all other inputs were kept the same including SEED values). A Monte Carlo simulation design was defined for the selection of mean threshold pressures for each shale layer with a range defined as 50 % above and below the deterministic mean (i.e. minimum of 49 kPa and maximum of 147 kPa). For each run, the sampled mean in each shale layer was used to define their corresponding normal distribution allowing each shale layer to vary independently from each other before running the realisation into the IP simulator. If we assume the permeability-threshold pressure transform proposed by [Nhabanga and Ringrose \(2022\)](#) this input range of mean threshold pressures for the shales corresponds to 0.08 to 1.5 mD of vertical permeability (assumed anisotropy in the shales is 0.01).

[Furre et al. \(2019\)](#) have identified several breaches or breaks in the intraformational shales that act as vertical conduits (also referred to as feeders) controlling the CO₂ migration which could be caused by faults, sand injectites or erosional features. These interpreted feeders observed in time-lapse seismic were included in the CO₂ Datashare Dataset as polygons. Therefore, an alternative scenario was considered with inclusion of these breaches in the intraformational shales in the 3D model which are hereby referred to as "shales with breaks" scenario or "SB". The initial scenario with laterally continuous shales is denoted as "continuous shales" or "CS". These polygons define three breaks in the shales by 2010: a main feeder (also called a chimney) that intersects all the intrashale layers and is located about 65 m to the south of the injection point (approximately 100 m × 100 m laterally); a North-Eastern feeder from L5 to L6 in which one cell (i.e. 50 m × 50 m) was selected

randomly from the given low confidence area polygon; a South-Western feeder from L7 to L8 in which also one cell was selected randomly from the low confidence area provided (Fig. 5). All feeders have been assigned the same properties as the sands. No uncertainty in the selection of the North-Eastern feeder and South-Western feeder within the low confidence polygons was considered. The same threshold pressures defined for the CS scenario were applied to the SB scenario, both in the deterministic model, with same the same normal distribution in all shale layers, and in the input range for the sensitivity analysis, which allows the shale layers to have independent distributions from each other. Note that recent work has identified several more vertical feeders at Sleipner (Furre et al. 2024, Martinez et al., 2024) so that the feeders modelled in this study are likely a subset of the true number of such features.

The values of mass distribution per layer in 2010 used as a reference in this study are shown in Table 1. They are based on the estimated area defined by the seismic interpreted polygons of each CO₂ plume layer extent in 2010, available from the CO₂ Datashare, which are shown in Fig. 6. Previous work (Furre et al., 2024) has shown a close correlation between 4D seismic polygon area and injected mass, although there are some uncertainties in this estimate.

4. Results

Based on the results of the sensitivity assessment of the intrashales threshold pressures, we picked out a selected case for each scenario that best matched the reference estimation of mass accumulation per layer in 2010, and studied the distribution range of CO₂ mass accumulation (Mt) per layer for all 500 realisations per scenario, along with their accumulation frequencies (%) through time.

Fig. 7 shows the CO₂ mass distribution (%) per layer in 2010 comparing the deterministic base case (initial model with same threshold pressure distribution in all shale layers), the selected case (realisation from the sensitivity study that best matched the reference values) and the reference case ("true values" estimation based on the areal extents of the plumes as shown in Table 1) for each scenario. Appendix B shows their corresponding CO₂ mass distribution in Mt per layer in 2010. The results demonstrate that the selected case for the CS

Table 1

Proportion of accumulated mass of CO₂ per layer estimated based on areal extent of injected CO₂ plumes by 2010 used as reference values (Callioli Santi et al., 2022).

Reference values (2010)		
Layer	Area (km ²)	Estimated percentage (%)
L9	2.13	17
L8	1.32	11
L7	0.84	7
L6	0.95	8
L5	3.02	25
L4	0.78	6
L3	0.88	7
L2	1.25	10
L1	1.11	9

scenario gives an excellent match per layer, except for L5, where the selected case shows 21 % compared to 25 % in the reference for this layer. For the SB scenario, the overall match is good but there is more CO₂ in L9 than in the reference case. Also, L2 and L3 show no significant variation in mass accumulation for both cases (the deterministic base case and selected case have approximately 1 % CO₂ in these layers vs the reference with 10 % in L2 and 7 % in L3). This is because the structural highs in these layers in the 3D model coincide with the main feeder. Therefore, most of the CO₂ bypassed these two layers through the main feeder accumulating in shallower layers in all realisations. Table 2 shows the root square mean error (RMSE) of the mass distribution for each selected case indicating that the other layers in the SB scenario have a good match to the reference case.

Fig. 8 shows the CO₂ mass distribution of the selected cases in 3D view. It shows that the CO₂ plume flows up dip from the injection point and the nine accumulations conform to the topography of each shale layer. The 3D grid presents a structural high to the south of the injection point in the deepest L1. In the CS scenario, the CO₂ migration pattern is restrained to filling and breaching the south structural closures until it reaches topmost L9 and spills laterally to the north below the caprock. In the SB scenario, the main feeder controls the vertical migration of CO₂ in

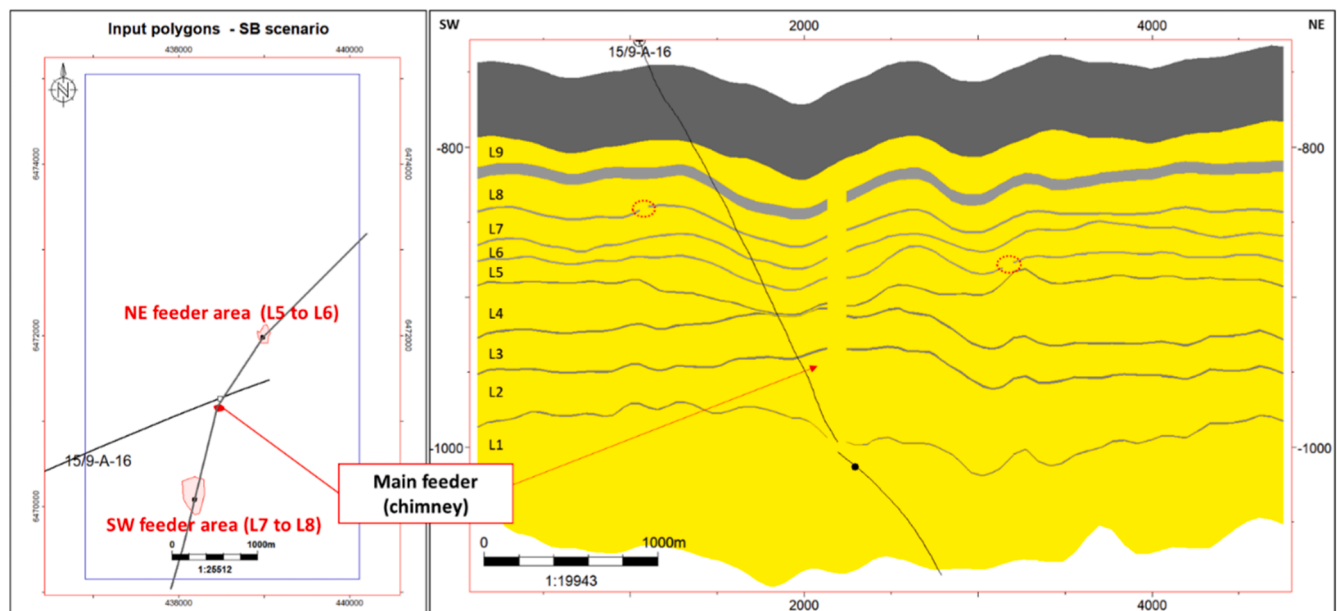


Fig. 5. Left: Map view of the input breaks in shales polygons from the CO₂ Datashare in red, each selected cell within the low confidence areas in black, the intersection line in dark grey and the blue polygon showing 3D model boundary. Right: Intersection roughly SW-NE displaying the breaks in the shales in the 3D model, the injector well 15/9-A-16 and the injection point (black circle) (VE 9x). Main feeder (chimney) going through all the shale layers, one cell in shale layer 7 towards the SW and one cell in shale layer 5 towards the NE, both highlighted with dotted red circles. Yellow shows Utsira sandstones, light grey shales and dark grey above L9 is the caprock (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

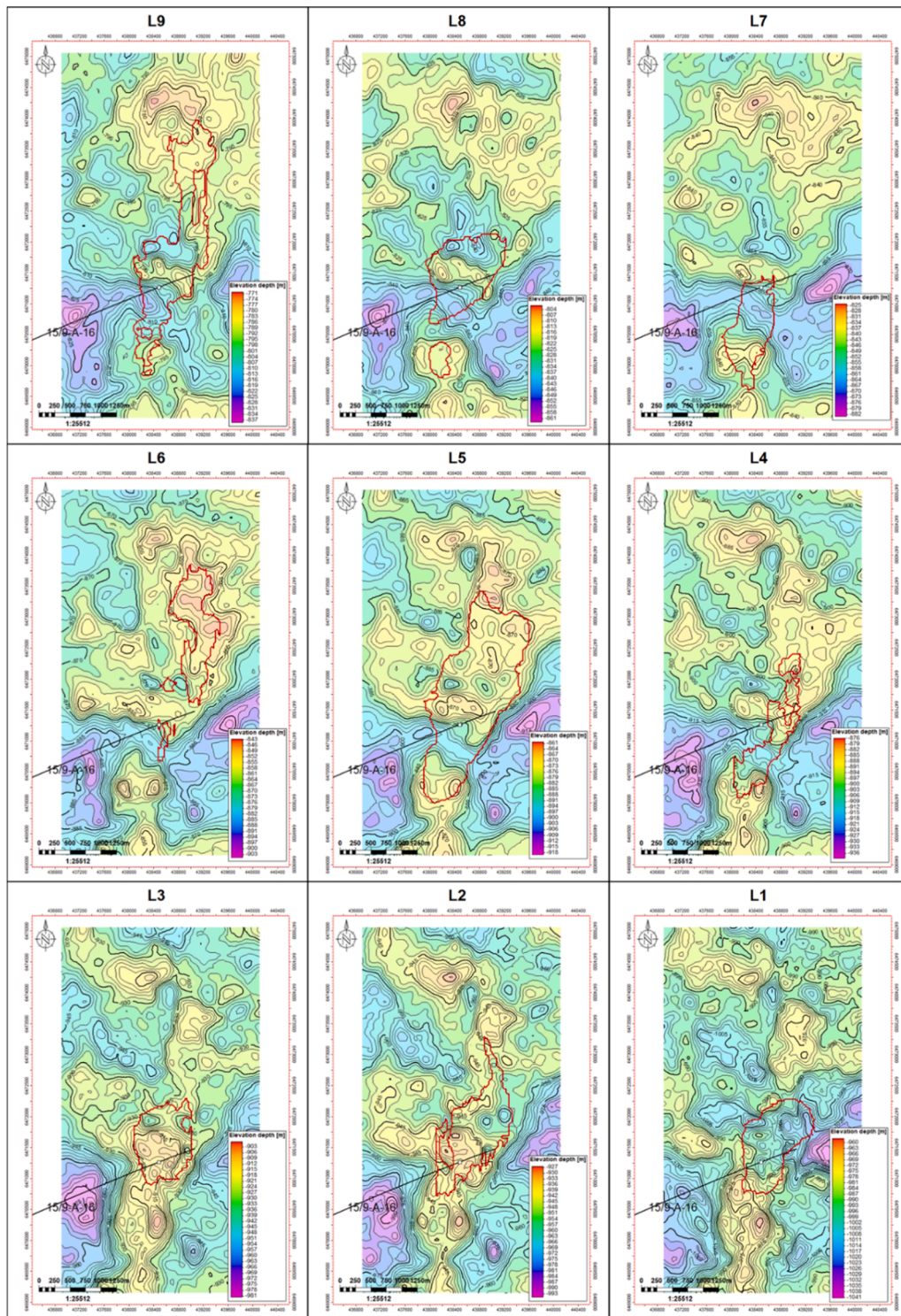


Fig. 6. CO₂ plume outlines (boundary polygons) for each layer mapped from seismic time-lapse data in 2010 are shown in dark red overlaying their corresponding top structural map from the Sleipner 2019 Benchmark mode (L1 is deepest and L9 shallowest) (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

layers 2 to 4 which moves the plume towards the north with more lateral migration in the middle layers 4 to 6. The North-Eastern feeder controls the CO₂ upward migration from L5 to L6. The migration towards the north moves the CO₂ away from the main feeder in the layers above L4 so the South-Western feeder was not reached between L7 and L8. Considering observations from seismic (as illustrated in Figs. 3 and 6), the SB scenario presents a better overall match to the lateral distribution

of the CO₂ plume than the CS scenario.

The distribution of threshold pressures in the shale layers in the selected cases is summarised in Table 3. Note that the CS scenario shows the highest mean value of threshold pressure in the thick shale (above L8) while the smallest is in shale 7 (above L7). The SB case resulted in the largest mean value in shale 2 (above L2) and the smallest also in shale 7 (above L7). In the SB scenario, the vertical migration of CO₂ is

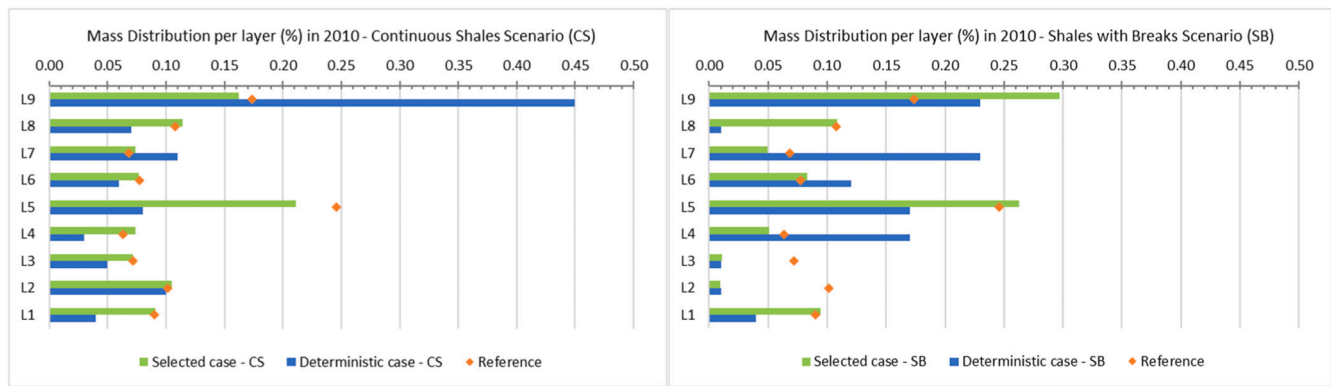


Fig. 7. Left: Distribution of CO₂ mass in percentage per layer in 2010 for the deterministic base case, the selected case and the reference estimation for the CS scenario. Right: Distribution of CO₂ mass in percentage per layer in 2010 for the deterministic base case, the selected case and the reference estimation for the SB scenario.

Table 2

Root Mean Square Error (RMSE) of the mass distribution for each selected case (CS and SB).

Layers	Selected case - CS Squared Difference	Selected case - SB Squared Difference
L9	0.02	2.28
L8	0.01	0.00
L7	0.00	0.05
L6	0.00	0.00
L5	0.17	0.04
L4	0.02	0.02
L3	0.00	0.55
L2	0.00	1.25
L1	0.00	0.00
RMSE	0.16	0.68

mostly controlled by the feeders in shales 2 (L2 to L3), 3 (L3 to L4) and 5 (L5 to L6).

The resulting capillary pressure of the CO₂ (due to buoyancy pressure) and the maximum CO₂ column heights (maximum height above spill point) per layer for each selected case are shown in Table 4. In L1 the distribution is almost identical in both cases as the CO₂ migrates southwards, so the main feeder plays no role in this layer in the SB scenario. From L2 to L8 the buoyancy pressures of the CO₂ are consistently higher in the CS scenario. This is due to the CO₂ mostly accumulating in the South structure in the CS scenario which shows overall larger heights above spill points than the closures to the north. L9 (below the caprock) shows a higher buoyancy pressure in the SB scenario as more CO₂ has reached this layer than in the CS.

The L6 mass distribution in both cases (and in the reference case) corresponds to 8 % of the total mass in 2010 (Fig. 7 and Table 1). However, the CO₂ upward migration in the CS scenario is restricted to the south of the 3D model as shown in Fig. 8. Fig. 9 shows a closure and fetch analysis of the top structural map of L6 and the capillary fluid pressures (CO₂ buoyancy) in both the CS and the SB selected cases. We observe that the structural closure towards the south has a height above spill point of 19 m in L6. The CS selected case reaches a maximum CO₂ buoyancy pressure of 211 kPa, while the mean threshold pressure for shale 6 in this case is 112 kPa (Table 3). It is also observed that the structural closure towards the north in L6 defines a larger area with gentler slope and height above spill point of 16 m. Consequently, in the SB selected case, the CO₂ buoyancy pressure reaches a maximum of 134 kPa, and shale 6 has a mean threshold pressure of 86 kPa. Therefore, although shale 6 has lower threshold pressures in the SB selected case, the filling of structural traps towards the north (driven by upward migration from L5 via the North-Eastern feeder) leads to a greater lateral spread of CO₂ before it breaches shale 6 and migrates to L7. This

interplay gives nearly the same mass accumulation as in the CS selected case.

Table 5 shows the summary of CO₂ mass distribution (Mt and %) per layer in 2010 from the 500 realisations of threshold pressures for each scenario (CS and SB). We start by looking at the results from L5 and L9 which are the largest accumulations in 2010 (reference values of 25 % and 17 % respectively as shown in Table 1). The CS scenario shows that in L5 the selected case is close to the P10 whereas the deterministic base case is close to the P50 (Fig. 10). Because the selected case has the closest match to the estimated reference mass balance, the input range distribution of threshold pressure in this shale layer is likely too low. In the SB scenario, the results from the sensitivity analysis for shale L5 also shows that the selected case corresponds to the P10 and the deterministic base case corresponds to the P50. In this scenario, the CO₂ from L5 migrates vertically through the North-Eastern feeder and moves up dip filling the structural trap towards the north in 93 % of the realisations.

The sensitivity results for L9 show similar mean values for the mass in both scenarios (Table 5) but the P50 for the CS gives 2.46 Mt of CO₂ accumulated in L9 in 2010 (corresponding to 20 % of the total mass injected by 2010) whereas the P50 for the SB is 0.61 Mt (corresponding to 5 % of the total mass injected by 2010). Therefore, the distribution of mass in L9 in the SB scenario are skewed towards low values with the selected case at P28 and the deterministic base case at P34 as illustrated in Fig. 11. The selected case for the CS corresponds to a P54 with 1.98 Mt accumulated in L9 (by 2010) and the deterministic base case corresponds to a P16.

Looking at the distribution in the other layers (Table 5), in the CS scenario, L8 shows a distribution towards low values with a P50 CO₂ mass of 5 % and a P10 of 14 % (vs 11 % in the reference) which indicates that the threshold pressure range for this layer could also be too low. In the SB scenario, the CO₂ preferentially migrates through the main feeder in layers 2 and 3 with little variation in the CO₂ accumulations in these two layers in the results from the sensitivity analysis (500 realisations). L4 shows a mass distribution in 2010 with P50 of 17 % whereas the reference is 6 %; and L6 gives a P50 of CO₂ mass distribution in 2010 of 12 % vs reference 8 %. For L8, the CO₂ mass distribution in 2010 has a P50 value of 1 % and a P10 of 10 % (vs 11 % in the reference). These results show that vertical feeders and the topography are also important factors contributing to increase the lateral spread of the CO₂ plume in layers 4 to 6, which then reduced the mass distribution of CO₂ in the upper layers 8 and 9.

Fig. 12 shows a north-south cross-section of the CO₂ accumulation frequency from the 500 realisations for the CS scenario in 1999 and 2010, and Fig. 13 for the SB scenario. These are also displayed in 3D view in Appendix B. The first time-lapse seismic survey acquired in Sleipner in 1999 already showed small accumulations below the caprock (Zweigel et al., 2004). The accumulation frequency result for the CS

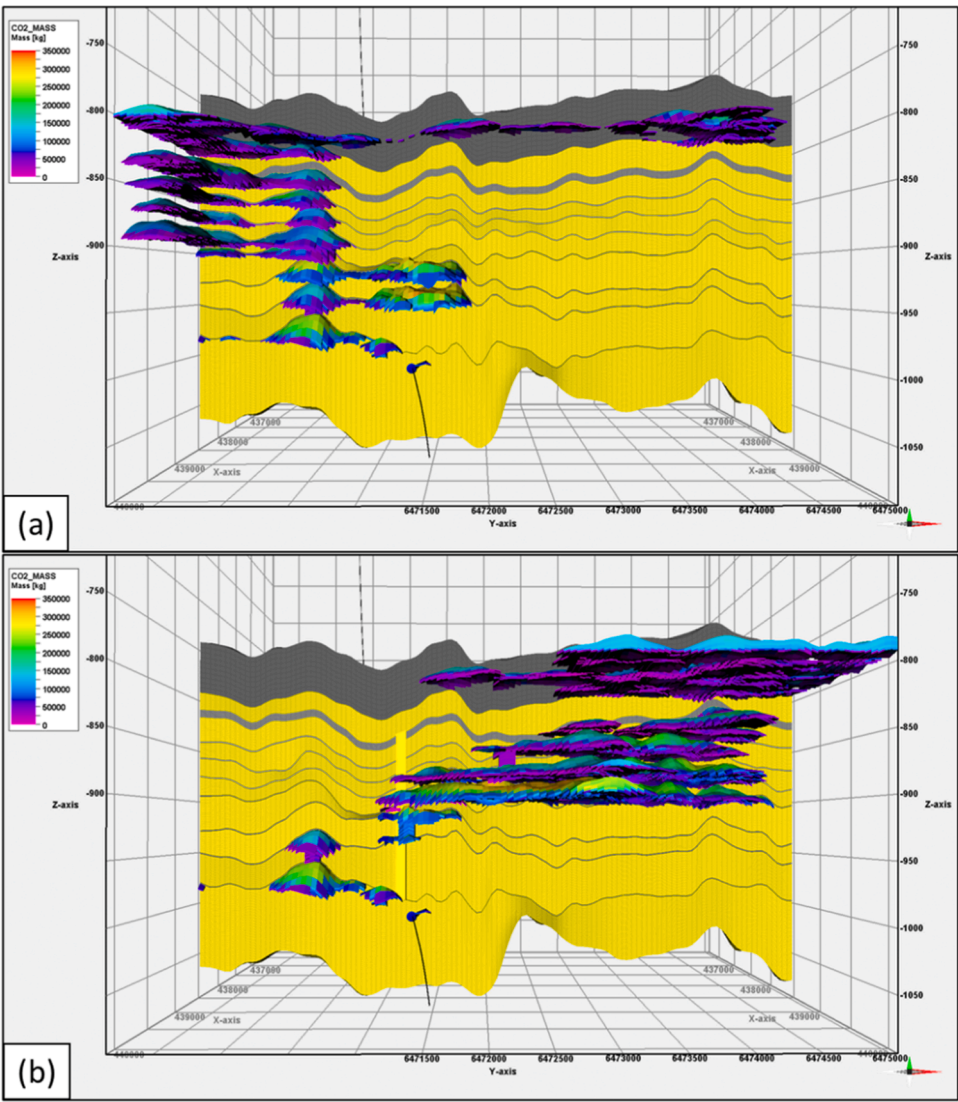


Fig. 8. CO₂ Mass distribution in 3D view (View from East – compass bottom right: red points north and green points up, VE 10x). (a) Selected continuous shale case (CS). (b) Selected shales with breaks case (SB).

Table 3

Threshold pressures distributions per shale layer in the selected cases (CS and SB). *Excluding the main feeder and the NE and SW feeders as these have the same properties as the Utsira sandstones. In italic are the shale layers where the feeders controlled the vertical migration of CO₂ (shales 2, 3 and 5).

Shale layers	Shale layers P _{th} (kPa)			
	Selected case - CS		Selected case - SB*	
	μ	σ	μ	σ
Thick shale	141	14	87	8
Shale 7	80	8	76	7
Shale 6	112	11	86	8
Shale 5	136	13	101	10
Shale 4	122	12	118	11
Shale 3	100	10	119	12
Shale 2	95	9	138	13
Shale 1	117	11	118	11

scenario showed that the earliest that the CO₂ reached L9 was in 2000 (one realisation out of 500 or 0.2 % of the realisations as shown in Table 6). After 7 years of injection, only 13 % of the realisations reached L9 and 68 % reached L5. By 2010, 72 % of the realisations reached L9 and 95 % reached L5. This shows that the threshold pressures of the

Table 4

Modelled CO₂ buoyancy pressure and maximum CO₂ column heights (maximum height above spill point) per layer for the selected cases from the sensitivity study (CS and SB).

Plume layers	CO ₂ buoyancy pressure (kPa)						Maximum CO ₂ column height (m)	
	Selected case - CS			Selected case - SB			Selected case - CS	Selected case - SB
	μ	σ	Max	μ	σ	Max		
L9	44	36	167	54	45	247	17	23
L8	39	41	192	17	28	183	20	11
L7	36	31	134	26	23	134	14	14
L6	48	39	211	40	31	134	22	14
L5	76	52	238	41	33	183	24	18
L4	49	42	190	24	21	110	20	21
L3	33	35	175	22	18	91	18	9
L2	46	38	183	37	34	140	19	14
L1	57	45	229	57	46	231	23	24

laterally continuous shales do play an important role in limiting the upward migration of CO₂ from the injection point to the caprock.

The 500 realisations for the SB scenario resulted in CO₂ first reaching L9 by 1999 (one realisation as illustrated in Fig. 13). By 2003, 9 % of the

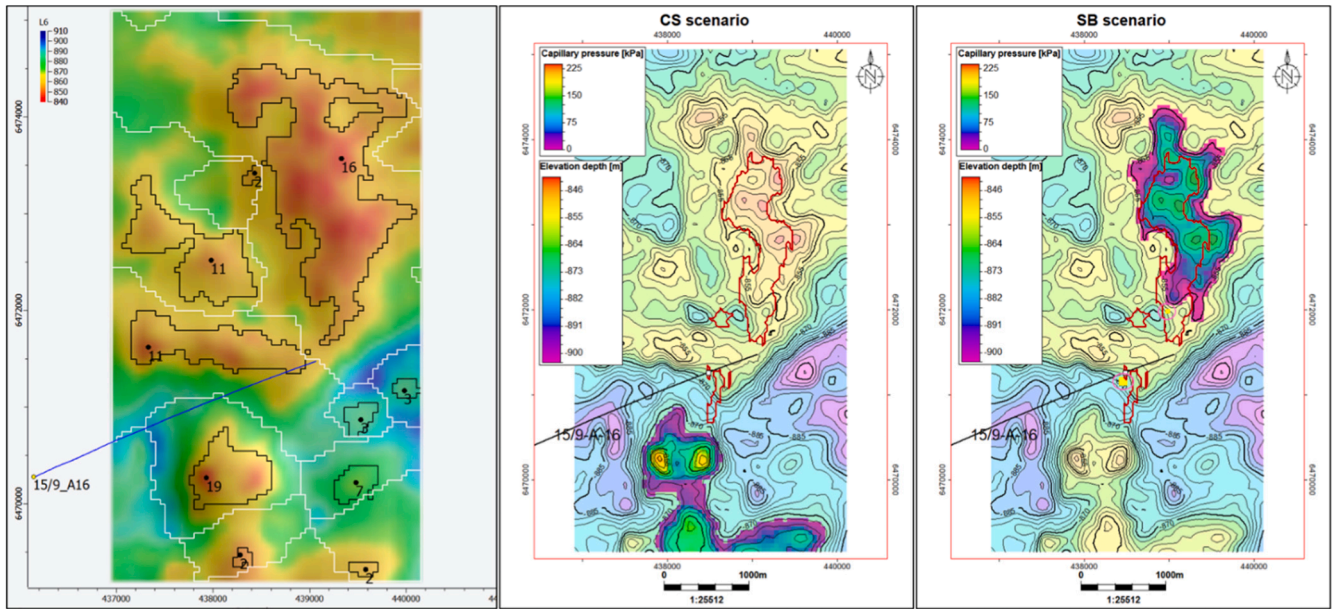


Fig. 9. Left: Top Reflector 6 (L6) closure and fetch analysis (defined 10 largest closures by area). Closures polygons are shown in black, fetch-area polygons are shown in white, black circle points show maximum closure heights (m). Middle: Capillary fluid pressures (kPa) at Top Reflector 6 (L6) for the CS selected case. Dark red polygon shows observed plume outline in 2010. Right: Capillary fluid pressures (kPa) at Top Reflector 6 (L6) for the shales for the SB selected case. The main feeder and the North-Eastern feeder from L5 to L6 (yellow squares) are highlighted with pink circles. Dark red polygon shows observed L6 plume outline in 2010 (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

Table 5

CO₂ mass distribution per layer in 2010 (total of 12.18 Mt of CO₂ injection) for the 500 realisations of threshold pressures for each scenario (CS and SB).

Intrashales threshold pressures sensitivity study – continuous shales scenario (CS)														
Plume layer	Mass distribution (Mt) - 2010							Mass distribution (%) - 2010						
	μ	σ	Min	Max	P50	P90	P10	μ	σ	Min	Max	P50	P90	P10
L9	2.70	2.48	0.00	9.17	2.46	0.00	6.38	22	20	0	75	20	0	52
L8	0.74	0.66	0.00	2.42	0.61	0.00	1.76	6	5	0	20	5	0	14
L7	1.35	1.12	0.00	3.56	1.16	0.00	3.03	11	9	0	29	10	0	25
L6	0.78	0.67	0.00	2.31	0.60	0.00	1.88	6	5	0	19	5	0	15
L5	1.22	0.90	0.00	3.10	0.97	0.26	2.63	10	7	0	25	8	2	22
L4	0.61	0.54	0.00	1.97	0.47	0.04	1.58	5	4	0	16	4	0.4	13
L3	1.02	0.85	0.00	3.29	0.75	0.09	2.30	8	7	0	27	6	1	19
L2	2.86	3.21	0.05	10.81	1.35	0.06	8.51	24	26	0.4	89	11	1	70
L1	0.76	0.53	0.14	1.96	0.60	0.21	1.62	6	4	1	16	5	2	13
Intrashales threshold pressures sensitivity study – shales with breaks scenario (SB)														
Plume layer	Mass distribution (Mt) - 2010							Mass distribution (%) - 2010						
	μ	σ	Min	Max	P50	P90	P10	μ	σ	Min	Max	P50	P90	P10
L9	2.07	2.60	0.00	10.05	0.61	0.00	6.23	17	21	0	82	5	0	51
L8	0.33	0.52	0.00	2.20	0.08	0.00	1.22	3	4	0	18	1	0	10
L7	1.75	1.97	0.00	8.96	0.89	0.00	4.64	14	16	0	74	7	0	38
L6	2.08	1.80	0.00	7.24	1.47	0.36	4.89	17	15	0	59	12	3	40
L5	2.08	1.22	0.08	3.20	2.65	0.19	3.20	17	10	1	26	22	2	26
L4	2.76	2.10	0.04	8.31	2.10	0.62	6.30	23	17	0	68	17	5	52
L3	0.13	0.02	0.05	0.13	0.13	0.12	0.13	1	0	0	1	1	1	1
L2	0.11	0.02	0.05	0.17	0.12	0.06	0.12	1	0	0.4	1	1	1	1
L1	0.76	0.53	0.14	1.96	0.60	0.21	1.62	6	4	1	16	5	2	13

realisations reached L9 and 69 % reached L5 as shown in Table 6. In 2010, 54 % of the realisations showed accumulations in L9 and 84 % in L5. This result indicates that in the SB scenario the CO₂ reached the caprock faster than in the CS scenario (similar to that indicated by the observations in the time-lapse seismic). Also, note that the SB scenario constrains the vertical migration of CO₂ to L9 even more than for the CS scenario.

Note that these results are highly dependent on the details of the structural 3D grid (especially the modelled depths) and the version of feeders included in this study. There are, however, still significant uncertainties in the modelled deeper layers of the 3D grid and we know that the included vertical feeders are likely a subset of reality. An

improved match to the observations could potentially be achieved by further work using updated data on vertical feeders (e.g. Furre et al., 2024; Martinez et al., 2024). Nevertheless, our model results give important insights on the interplay between threshold pressures of the shales, topography and vertical conduits which can significantly restrain and redirect the CO₂ migration in the storage unit.

5. Discussion

We have shown how the IPMC approach can be used to model vertical migration in a multi-layer system, both conceptually and applied to the real case of the Sleipner 3D model. Migration through a low

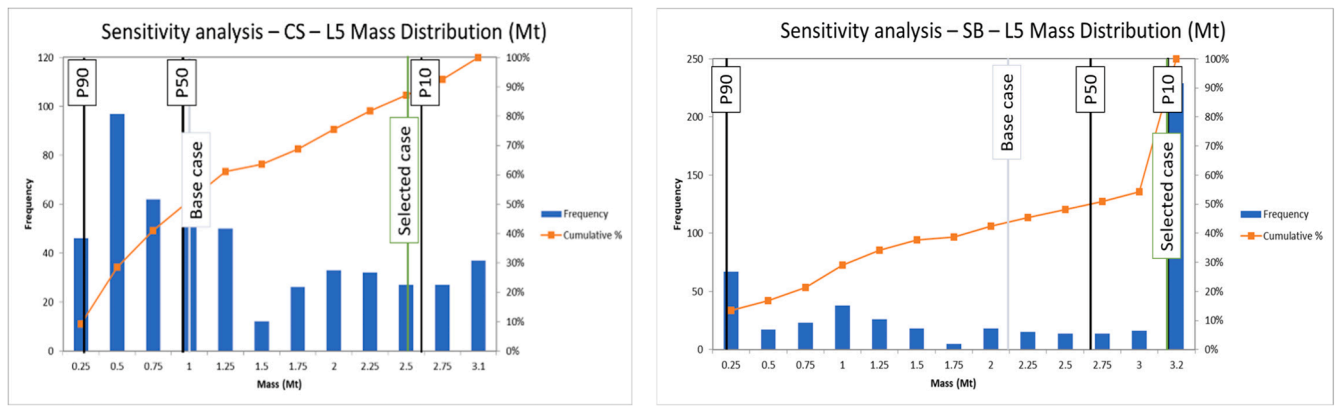


Fig. 10. Left: Distribution of CO₂ mass accumulation (Mt) in layer 5 from the sensitivity analysis (500 realisations) for the CS scenario. Right: Distribution of CO₂ mass accumulation (Mt) in layer 5 from the sensitivity analysis (500 realisations) for the SB scenario.

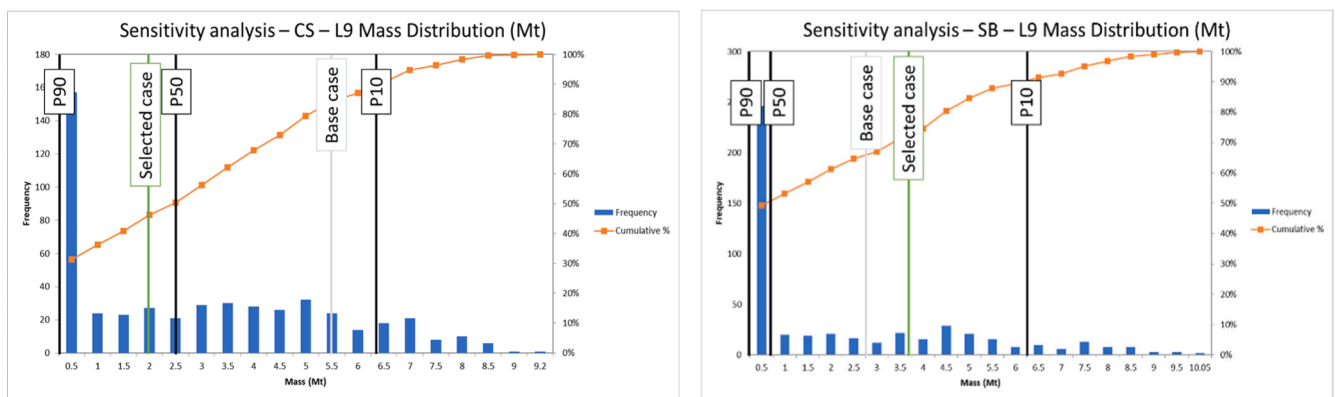


Fig. 11. Left: Distribution of CO₂ mass accumulation (Mt) in the topmost layer 9 from the sensitivity analysis (500 realisations) for the CS scenario. Right: Distribution of CO₂ mass accumulation (Mt) in the topmost L9 from the sensitivity analysis (500 realisations) for the SB scenario.

permeability layer cannot occur without a sufficient CO₂ volume accumulation in the permeable rock unit immediately below and is dependent on the geometries and capillary threshold pressures of the deeper layer (Fig. 1). Pathways with low threshold pressures within the sealing units such as faults, injection features or sand-to-sand contacts due to erosion allow the CO₂ to migrate vertically if these are located up dip or within a structural trap.

The IP simulation approach applied to the Sleipner 3D model confirms that the dominantly vertical migration of CO₂ follows a sequence of events in which migration into a shallower layer is directly dependent on the layer immediately below, supporting a Markovian model. The amounts of injected CO₂ volumes immobilised in each layer depend on the geometry of the traps (topography), the threshold pressures in each shale layer and on the presence of shale breaks.

In the demonstration of the IPMC method for the Sleipner case study, we obtained a good match to the CO₂ mass distributions per layer in 2010 for the continuous shale (CS) scenario (Table 2). The shale with breaks (SB) scenario gave a poorer match to the observed mass distribution per layer but a better spatial distribution, revealing how vertical pathways can have a major impact.

The CS scenario corresponds to the *a priori* knowledge set for the Sleipner case study with the assumption of continuous shale layers similarly to the analytical approach discussed in Section 3.2. The probability distribution of CO₂ mass per layer in 2010 from the sensitivity assessment of threshold pressures of the shales in this scenario, gave a P50 mass in L9 of 20 % (cf. reference value of 17 %, Table 5) and the selected case gave a P54 outcome in L9 (Fig. 11). This demonstrates that the IPMC approach for the CS scenario provided a good estimate of

the distribution of CO₂ mass in L9 with the input range of threshold pressures in the shale layers defined in the sensitivity study. Furthermore, the selected case for this scenario achieved a mass balance within a few percent of the estimated true values for most layers (Fig. 7 and Table 2). Considering the probability distribution of CO₂ mass in 2010 in the lower layers, this approach also provided a good estimate overall, but for the shale layers 5 and 8 the input distributions for threshold pressures would need to be increased to improve the match to the observations, as there is more than 80 % probability that the mass distribution in these layers is less than the reference.

The distribution of CO₂ accumulations per layer also conformed to the geometry of the traps (3D structural grid) in both scenarios. However, the lateral distribution of the CO₂ plume accumulations in the lower layers 1 to 8 in the CS scenario was restricted to the southern part of the 3D grid, unlike the observations from 4D seismic (Figs. 3 and 6). The sequence of CO₂ migration events starts with CO₂ migrating up dip towards the south from the injector filling and breaching a southern structural closure in the deepest layer (L1) in all 500 realisations in both scenarios (as shown in Fig. 12 for the CS scenario and in Fig. 13 for the SB scenario). Layer 4 nearly pinches out near the injector at the location of the main feeder and this reduction in thickness impacts the CO₂ lateral migration from the main feeder (i) towards the south in the SB scenario and (ii) from the South structural high towards the north in the CS scenario. Small variations in the estimated depth surfaces can significantly impact the migration of CO₂ because the column heights play a significant role in determining the pressure needed to breach the shales. The topography of the deeper layers 1 to 4 represent the largest uncertainties in the model build of the Sleipner 2019 Benchmark 3D

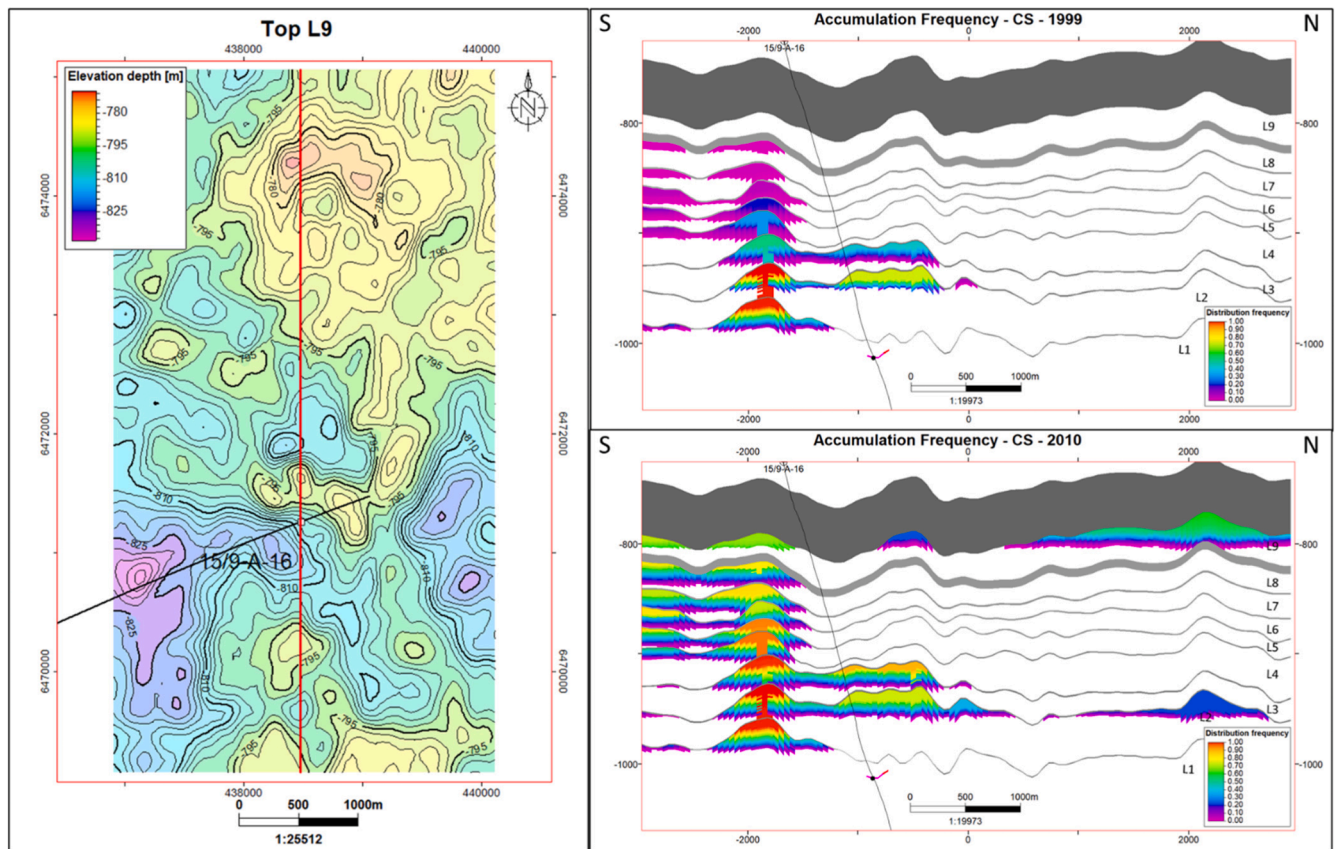


Fig. 12. Left: Map view showing N-S cross section in red. Top right: Accumulation frequency from the sensitivity study (500 realisations) for the CS scenario in 1999. Bottom right: Accumulation frequency from the sensitivity study (500 realisations) for the CS scenario in 2010. Purple colours show accumulation frequency of 0.1 and below. Red colour shows accumulation in all 500 realisations (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

model (as documented in the CO₂ Datashare portal) and these deeper layer assumptions determine the subsequent migration into the shallower layers 5 to 9. Therefore, it is a considerable challenge to match all nine layers of the observed plume distribution in Sleipner with a single realisation of the structural 3D grid. This observation highlights the need for a probabilistic analysis of the structural model at Sleipner in future studies, particularly in the deep layers 1 to 4.

Given the structural model defined by the Sleipner 2019 Benchmark, the observed lateral distribution of the plume can only be captured with the inclusion of breaks in the shales which allows the CO₂ to migrate northwards as demonstrated by the SB scenario (Figs. 8 and 13). However, the probability distribution of CO₂ mass in L9 from the sensitivity assessment for the SB scenario gave a P50 of 5 % (Table 5) – much less than observed. Thus, although the SB scenario demonstrates how the CO₂ preferentially migrates through vertical conduits that break up the shale barriers and gives an improved forecast of the lateral distribution of the CO₂ plume, it does not give a good mass balance forecast. The probability distributions of CO₂ mass in 2010 (SB scenario) were skewed towards low values in the topmost layers 8 and 9 and towards high values in middle layers 4 and 6. A better mass balance was obtained for the selected case (SB scenario) with some divergence in layers 2, 3 and 9 (Fig. 7 and Table 2).

These findings from the sensitivity analysis for the SB scenario are controlled by a specific version of shale breaks used in this study (which we suspect is incomplete). We observe that most CO₂ bypasses L2 and L3 through the main feeder (in all 500 realisations) and that the CO₂ movement remains constrained to the north (due to the break in shale 5) with large amounts of CO₂ accumulating in layers 6 and 7 and less CO₂ reaching upper layers 8 and 9 overall (as shown in Table 5). The South-

Western feeder from L7 to L8 was not reached since the accumulations were constrained to the north of the grid (from L4 and above). For CO₂ to migrate towards the south (migration from L6 to L7 and from L7 to L8) as observed in 4D seismic, additional feeders would be needed in the southwest allowing communication from L4 to L8. There is recent evidence from time-lapse seismic (Furre et al., 2024) that L6 is not in contact with the main feeder and that L7 is fed through the main feeder directly from L5. These observations indicate that an improved model of the shale breaks (feeders) is essential to get a good match to the observations at Sleipner.

It is also evident from the results that the probability of CO₂ vertical migration and accumulation in each layer decreases upwards (Table 6) as established by the IPMC method. At the time of the first 4D seismic survey in 1999, the CS scenario did not predict any realisation with accumulation in L9 and only 31 % of the realisations reached L4 whereas in the SB scenario one realisation reached L9 and 93 % of the realisations showed accumulations in L4. This is due to the main feeder controlling the vertical migration in layers 2 to 4 in the SB scenario. By 2010 more than 80 % of the realisations reached L7 in both scenarios. This demonstrates that the probability of vertical migration of CO₂ under continuous injection increases with time as illustrated in Fig. 2.

In both scenarios (CS and SB) the probability distribution of CO₂ mass in topmost L9 presented a P10 of approximately 6 Mt (Table 5). Hence, even in a high-end case for the mass distribution in L9 in 2010 (i. e. realisations with low threshold pressures in the shales 1 to 8), about 50 % of the total injected CO₂ is still retained in lower layers 1 to 8. This finding demonstrates the inherent value of a multiple seal system for the geological storage of CO₂. It also illustrates the features of a composite confining system (as proposed by Bump et al. 2023) in which minor

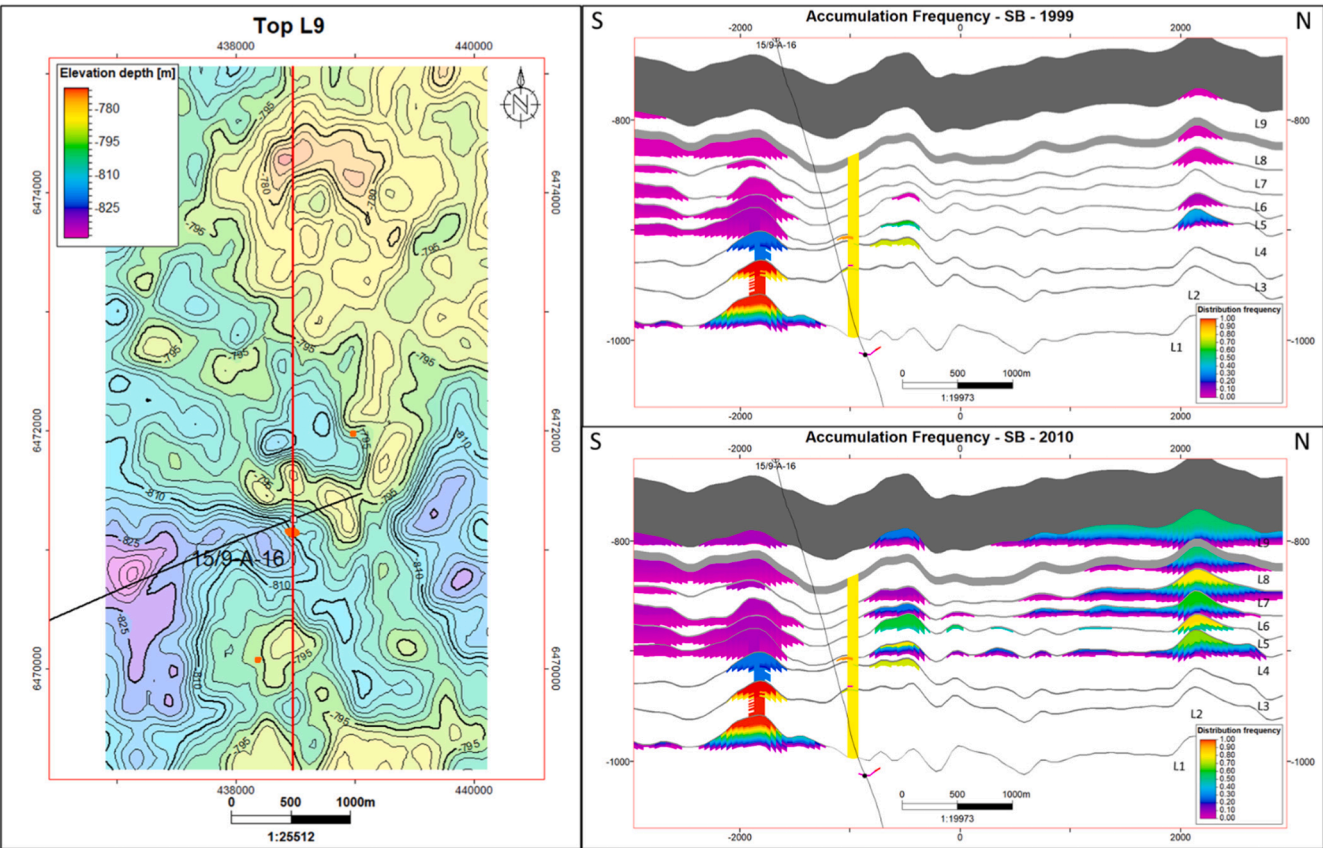


Fig. 13. Left: Map view showing N-S cross section in red, locations of the 3 feeders in orange. Top right: Accumulation frequency from the sensitivity study (500 realisations) for the SB scenario in 1999. Bottom right: Accumulation frequency from the sensitivity study (500 realisations) for the SB scenario in 2010. Purple colours show accumulation frequency of 0.1 and below. Red colour shows accumulation in all 500 realisations (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

Table 6
Accumulations frequency (%) per layer through time from the sensitivity study for each scenario (CS and SB). *Maximum value per layer.

Plume layer	Intrashales threshold pressures sensitivity study - accumulation frequency (%)*							
	Continuous shales scenario (CS)				Shales with breaks scenario (SB)			
	1999	2000	2003	2010	1999	2000	2003	2010
L9	-	0.2	13	72	0.2	1	9	54
L8	0.4	2	22	78	1	2	8	54
L7	3	7	45	84	1	3	24	80
L6	6	15	54	82	7	11	55	90
L5	20	37	68	95	21	42	69	84
L4	31	48	72	96	93	93	93	93
L3	52	62	78	99	93	93	93	93
L2	100	100	100	100	100	100	100	100
L1	100	100	100	100	100	100	100	100

threshold pressures for shale layers are enough to impact the initial flow of CO₂ and reduce the CO₂ vertical migration to shallower layers. Therefore, significant volumes of CO₂ can be immobilised in deeper layers in a multi-layered stratigraphic system. This segregation of the CO₂ plume into multiple accumulations also increases residual and dissolution trapping, further enhancing the immobilisation of CO₂ in the long-term.

In our work on IP simulations, various attempts were made in improving the forecasts for lateral distribution of the CO₂ plumes by adding permeability barriers to the model. First in the deepest L1 in order to avoid the CO₂ filling the South structure and then by adding more barriers upwards (layer by layer) in an attempt to limit the lateral

movement of CO₂ (to match observations from seismic). However, no satisfactory result was obtained by adding barriers alone due to the above-mentioned challenges in layers 6 and 7. Also, we observed that changes in the deeper layers significantly impact the layers above so that making arbitrary adjustments becomes futile. Furthermore, we appreciate that the IP method does not fully capture the flow physics of CO₂ migration over such a short-time scale (1996 to 2010) and especially where viscous forces play an important role close to the injection well and in deeper layers. [Nazarian and Furre \(2022\)](#) have successfully history matched the observed CO₂ plumes in the Utsira Fm. up to the time of the 2020 survey with a compositional modelling approach in Pflotran-OGS, using a combination of breaks in shales (feeders) and inclusion of low permeability barriers (added to the Sleipner 2019 Benchmark 3D model). In addition to the main feeder, they included two feeders in the southwest between L2 and L5 and L5 and L9 and one feeder in the northeast connecting L6 to L8. This shows that even with a full physics simulator there are still challenges when attempting to match the CO₂ plume distribution observed at Sleipner.

Future efforts to forecast the multi-layer CO₂ plume behaviour at Sleipner will need to capture both the geological detail (especially the layer topography, shale breaks and vertical feeders) and the flow physics. The structural setting of the Sleipner storage site (as captured in the 2019 Benchmark model) gives a roughly North-South elongated spread of the CO₂ plume, which is captured in our SB scenario and similar to the seismic observations ([Furre et al., 2019; Furre et al., 2024](#)). However, the modelling we performed did not capture potential effects of lateral heterogeneities (such as channelised features) within the Utsira Fm. and the feeders we included were assigned the same properties as sands. [Furre et al. \(2024\)](#) suggest that the Utsira Fm. represents

a confined submarine fan deposit with North-South trending channels which are clearly observed in the topmost layer 9 but also likely present in deeper layers. This implies a complex reservoir architecture with sand-to-sand contacts potentially resulting from erosion and amalgamation between channel sands or sediment remobilisation and vertical sand injection. The Furre et al. (2024) revised geological interpretation is also important to the understanding of vertical feeders including the main feeder which connects several CO₂ layers and other feeders which only connect individual layers. Recently, Martinez et al. (2024) have interpreted six migration pathways based on full-waveform inversion (FWI) including identification of faults in the deeper layers. The inclusion of these recent depositional and structural interpretations has the potential to significantly improve the future assessments of the flow processes at Sleipner.

Our IP simulation approach of the Sleipner case study was focused on understanding the role of the intraformational shales in controlling the vertical migration of CO₂ from the injection point to the caprock, and not on history matching the CO₂ plume. Assumptions on internal variations in sandstone properties (e.g. due to channels) were not evaluated but these could potentially improve the overall match to the observed plume distribution. While we did model a vertical trend in CO₂ density (a function of pressure and temperature), more complex density changes could occur (because the CO₂ is close to the critical point). The importance of modelling temperature gradients was demonstrated by Nazarian and Furre (2022). Another key parameter to consider in future studies is CO₂ saturation. In the IP method we used, the CO₂ saturation is either at the maximum CO₂ saturation within the accumulations (backfilled cells beneath a baffle or seal) which we assumed to be 70 % (Appendix A) or at residual CO₂ saturation (invaded cells outside the accumulations) which was defined as 20 % in our analysis.

6. Conclusions

We have demonstrated that CO₂ vertical migration in a multi-layer stratigraphic system is inherently a Markov process in which the probability of migration into each layer depends only on the layer directly below. The vertical flow behaviour can be described by IP methods with the CO₂ accumulating below low permeability layers and migrating upwards once the buoyancy pressures of the accumulations overcome the threshold pressures of the barriers or until the CO₂ encounters a high permeability conduit, either where the sealing layer has been breached or where up-dip lateral migration can occur, resulting in multiple accumulations.

In addition to the role of geological heterogeneities in restricting the CO₂ vertical migration, the interfacial tension between the CO₂ and brine is a key factor. Residual trapping acts to significantly reduce the available mass of CO₂ for migration into shallower units. Dissolution trapping, which was not included in this analysis, would also contribute to immobilise the CO₂ in the long-term. Our analysis highlights the value of a multiple seal system for geological storage of CO₂. These findings from a multi-layered storage unit (Sleipner case study) are directly applicable to multi-layer migration in general, including migration out of the storage unit and into the overburden.

The probabilistic IP simulation approach for modelling vertical CO₂ plume migration at Sleipner demonstrates that the CO₂ migration behaviour is controlled by variations in the threshold pressure distributions of each intraformational shale layer and strongly conforms to topography. It also gave satisfactory matches to the CO₂ mass distribution in each of the nine layers of the Utsira Fm. The results were very sensitive to the presence of sand-to-sand contacts or shale breaks, which influenced the CO₂ plume's migration northward in our SB scenario, consistent with observations from the time-lapse seismic data. The study confirmed that the probability of vertical CO₂ migration in each layer

decreases upwards from L1 to L9 as defined by the IPMC method. Additionally, it demonstrated that the probability of vertical migration through the nine layers increased with time during the injection period.

A detailed match to seismic observations using this approach was challenging to achieve since a match in each shallower layer depends on the match of the layer immediately below. The nature of the deeper layers (1 to 4) represents the largest uncertainties in the 3D model build and the assumptions in these deeper layers have important consequences for models of the shallower layers. Also, the version of shale breaks applied in this study is potentially a subset of reality and our results showed that additional feeders are needed to optimise the match. Seismic imaging of the deeper layers and the vertical feeders is very difficult, but they could be improved using full waveform inversion (Martinez et al., 2024). Improved modelling of structural uncertainties (particularly in the deeper layers) and updated data on vertical conduits (feeders) (e.g. Furre et al., 2024; Martinez et al., 2024) is recommended for future studies as these assumptions strongly impact the migration pattern of the CO₂ through the nine layers. Threshold pressure assumptions for each shale layer are also important, and our study has demonstrated the most likely ranges for capillary threshold pressures at Sleipner.

These challenges point to the inherent uncertainties involved when applying the IPMC approach, since the probability of later migration events is highly dependent of the probability of preceding events. Conversely, it demonstrates the need to focus on the initial migration events (e.g. when monitoring other sites) since these have the highest probability of occurring.

Funding sources

This work was funded by the Norwegian Research Council (NRC) via the 'Research-based Innovation' (SFI) Centre for Geophysical Forecasting (CGF) (grant no. 309960).

CRediT authorship contribution statement

Andrea Callioli Santi: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Philip Ringrose:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Jo Eidsvik:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Tor Andre Haugdahl:** Software, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The work was conducted as part of the Centre for Geophysical Forecasting (CGF), a joint enterprise funded by the Norwegian Research Council (NRC), industrial partners and the Norwegian University of Science and Technology (NTNU). We thank Equinor Energy AS and the Sleipner license partners LOTOS Exploration and Production Norge AS, Vår Energi ASA and KUFPEC Norway AS for use of the Sleipner CO₂ Reference dataset made available via the CO₂ Storage Data Consortium (CO₂datashare.org). We thank Halliburton for the academic license of the Permedia® software package and SLB for use of the Petrel™ software.

Appendix A. Assumptions for the IP simulation of the Sleipner nine-layer 3D model

The annual CO₂ injection rate is given in Table A. We assume the thermodynamics properties of pure CO₂ noting that the minor components of methane and BTX hydrocarbons tend to cancel each other's effects (Zweigel et al., 2004).

Initial model conditions and property ranges are taken from Singh et al. (2010). Temperature assumptions were taken from Cavanagh and Haszeldine (2014) and Cavanagh et al. (2015) with 35 °C at 800 m depth. From Singh et al. (2010) we have assumed a geothermal gradient of 35.6 °C/km which resulted in CO₂ densities ranges from approximately 350 kg/m³ in the uppermost L9 to 570 kg/m³ in the lowermost L1. The value assumed for residual trapping of CO₂ was 20 % and the irreducible water saturation was set to 30 % (guided by Reynolds and Krevor, 2015 and Jackson et al., 2018).

The Permedia® software was used for the IP simulations. In this software package the capillary pressure values must be input for Hg-Air and the software computes a conversion factor (based on pressure, temperature, salinity and density contrast) to convert the values to the *in situ* CO₂-brine system before running migration simulation.

Table A
Sleipner annual CO₂ injection up to 2010 (in Mt).

Sleipner	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
Per year	0.07	0.67	0.85	0.94	0.94	1.02	0.96	0.92	0.76	0.87	0.83	0.93	0.82	0.86	0.76
Total	0.07	0.74	1.59	2.52	3.46	4.48	5.45	6.37	7.12	7.99	8.82	9.74	10.57	11.42	12.18

Appendix B. Results from the IP simulation of the Sleipner nine-layer 3D model

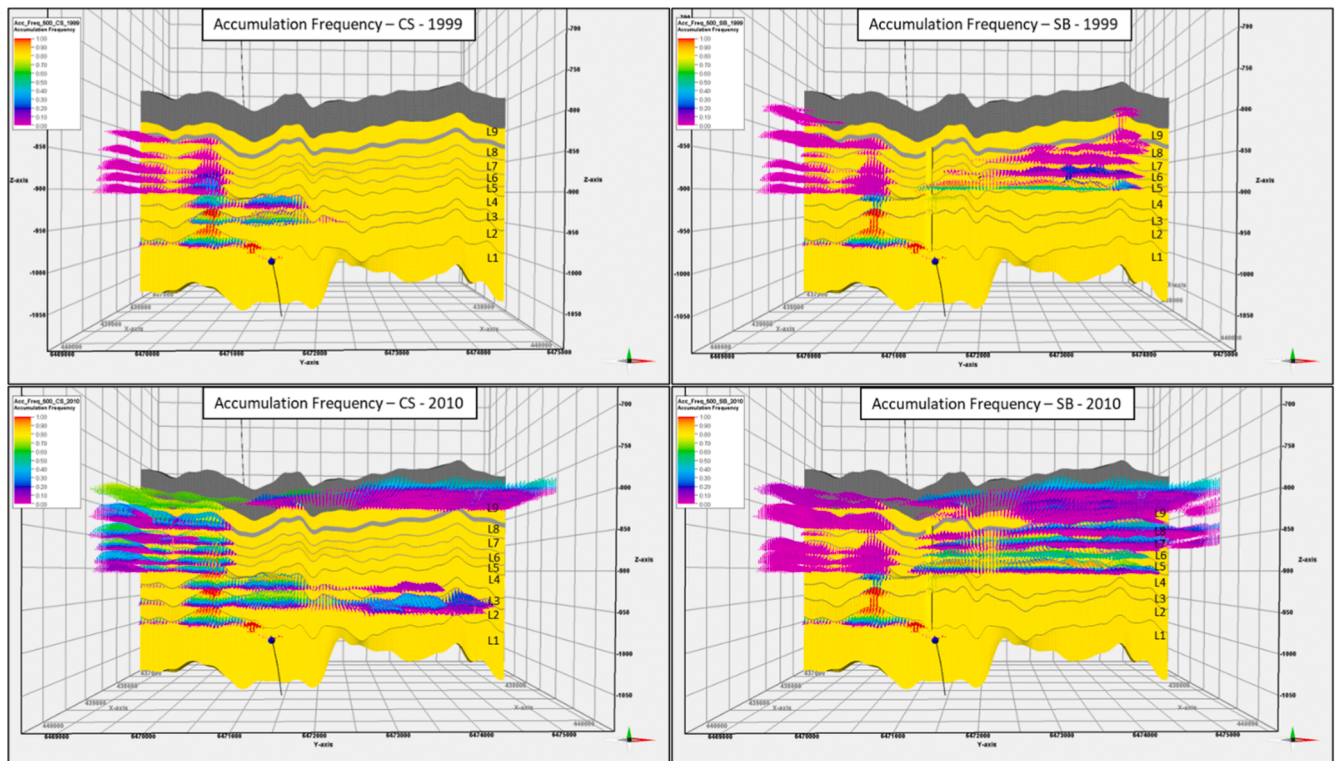
Table B gives the modelled CO₂ mass distribution (Mt) per layer in 2010 for the deterministic (same threshold pressures distribution in all shale layers) and selected cases (independent threshold pressures distributions in each shale layer that best matched the mass distribution per layer to the reference from the sensitivity study) for both scenarios, CS and SB. Fig. B shows 3D views of models for the CS and SB scenarios colour-coded by the CO₂ accumulation frequency based on 500 realisations for each scenario in 1999 and 2010.

Table B
CO₂ mass distribution per layer (Mt) by 2010 for the deterministic base case and the selected case for the continuous shales (CS) and the breaks in shales (SB) scenarios.

Layer	CO ₂ mass distribution (Mt) per layer in 2010			
	Deterministic base case - CS	Selected case - CS	Deterministic base case -SB	Selected case -SB
L9	5.49	1.98	2.77	3.62
L8	0.80	1.39	0.16	1.32
L7	1.28	0.90	2.77	0.60
L6	0.76	0.93	1.41	1.01
L5	1.01	2.58	2.10	3.20
L4	0.33	0.90	2.12	0.62
L3	0.63	0.88	0.13	0.13
L2	1.25	1.28	0.12	0.12
L1	0.48	1.11	0.48	1.15

Fig. B

Accumulation frequency from the sensitivity study (500 realisations for each scenario) in 3D view (view from East, VE 10x). Left: Accumulation frequency for the CS scenario (top shows in 1999 and bottom in 2010). Right: Accumulation frequency for the SB scenario (top shows in 1999 and bottom in 2010). Purple colours show accumulation frequency of 0.1 and below. Red colour shows accumulation in all 500 realisations (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).



Data availability

All work is reproducible by others using the open access Sleipner CO₂ Reference dataset available on the CO₂ DataShare portal (<https://co2datashare.org>).

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